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**Politecnico
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Nuclear Fission Plants

Assignments 1.1 & 1.2

Analysis and Design of Passive Decay Heat Removal Systems

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Chapter 1

Assignment 1.1 - Introduction to passive decay heat removal system mechanisms

1.1 Introduction

In this first assignment, the case study is a simple rectangular loop, where natural circulation is established through density gradients alone, driving the fluid around the loop. In this study, to simplify the analytical model of the system, it is assumed that both the heater and the cooler are positioned horizontally. The resulting physical mechanism is governed by a buoyancy-driven flow: saturated steam travels through the hot leg from the heater to the cooler, whereas saturated liquid recirculates through the cold leg, returning from the cooler to the heater to complete the closed-loop cycle.

The computational study is organized into three progressive stages:

1. Determine the vertical height h required to ensure that the specified thermal power is completely removed by natural circulation.
2. Assuming the horizontal leg length L as a design variable, calculate the equilibrium elevation under geometric constraint $h = L$, considering smooth pipes.
3. Re-evaluate the symmetric configuration ($h = L$) by incorporating the effect of pipe surface roughness, set at $\epsilon = 5.0 \times 10^{-5}$ m, on the overall hydraulic resistance.

1.1.1 Main assumptions

To streamline the mathematical formulation and focus on the primary thermal-hydraulic phenomena, the following simplifying assumptions are adopted for the natural circulation loop:

- **Geometric configuration:** Both the heater and the cooler are assumed to be horizontal, implying a constant elevation throughout these components.
- **Adiabatic legs:** Heat transfer between the fluid and the environment in both the hot and cold legs is neglected; therefore, the fluid properties change only within the heat exchangers.
- **Steady-state regime:** The system is analyzed under steady-state conditions, assuming time-invariant mass flow rates and temperature distributions.
- **Saturation conditions:** The working fluid is assumed to be at thermodynamic equilibrium at all times, with steam and liquid phases always maintained at saturation.
- **Uniform pressure:** A constant operating pressure of $p_l = p_v = 70$ bar is assumed for both the liquid and steam phases, neglecting local pressure variations for the evaluation of thermophysical properties.

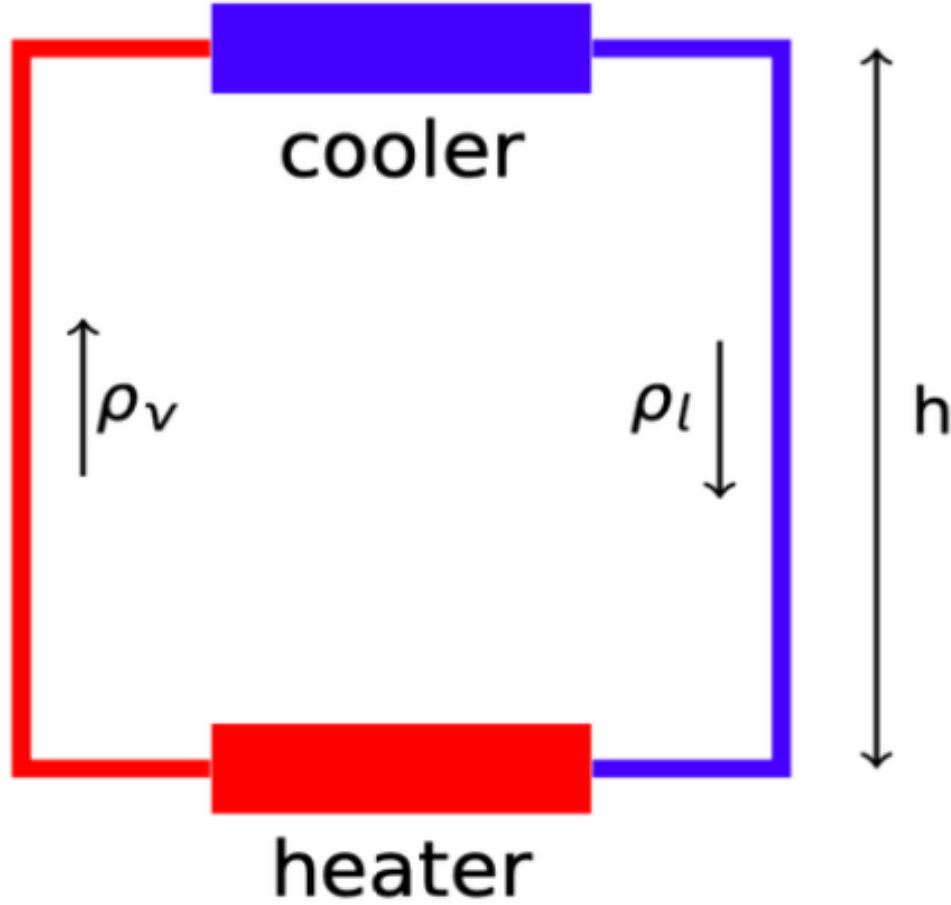


Figure 1.2: Natural circulation circuit diagram

1.1.2 Input data

The numerical simulation and the subsequent dimensioning of the loop are based on the operational and geometric specifications provided in the assignment. In accordance with the modeling requirements, the localized pressure drop coefficients are applied as follows:

- The coefficient $k_{cooler} = 20$ is associated with the velocity of the saturated liquid within the cooler component.
- The coefficient $k_{heater} = 40$ is associated with the velocity of the saturated steam within the heater component.

The pipe diameters used for the selection process are derived from standard commercial ANSI tables (NPS), considering the thickness associated with the "Standard" (STD) schedule.

1.2 Methodology to solve the assignment

The starting point is the momentum balance equation, which for the assumptions made reduces to:

Table 1.1: Operational and geometric parameters for the natural circulation loop.

Parameter	Symbol	Value
Total thermal power to be removed	P_{th}	34.8 MW _{th}
Length of each horizontal leg	L	10.0 m
Total horizontal circuit length	L_{total}	20.0 m
Elbow localized pressure drop coefficient	k_{elbow}	0.3
Cooler localized pressure drop coefficient	k_{cooler}	20.0
Heater localized pressure drop coefficient	k_{heater}	40.0
Operating pressure	p	70 bar
Acceleration of gravity	g	9.81 m/s ²
Pipe roughness	ϵ	5.0×10^{-5} m

$$\Delta p_{buoyancy} = \Delta p_{loss} \quad (1.1)$$

$$(\rho_l - \rho_v)gh = \Delta p_{distributed} + \Delta p_{concentrated} \quad (1.2)$$

$$(\rho_l - \rho_v)gh = \sum \left(f \frac{L}{D} \frac{1}{2} \rho v^2 \right) + \sum k \frac{1}{2} \rho v^2 \quad (1.3)$$

$$(\rho_l - \rho_v)gh = f_l \frac{L_l}{D} \frac{1}{2} \rho v_l^2 + k_{l,tot} \frac{1}{2} \rho v_l^2 + f_v \frac{L_v}{D} \frac{1}{2} \rho v_v^2 + k_{v,tot} \frac{1}{2} \rho v_v^2 \quad (1.4)$$

The heater and the cooler minor losses coefficients k_v and k_l are equal to 40 and 20, respectively. But if the four 90° fringed elbows are also considered (two in the steam side and two in the liquid side), the concentrated loss coefficients are evaluated as follows:

$$k_{v,tot} = K_v + 2 \cdot k_{elbow} = 40 + 2 \cdot 0.3 = 40.6 \quad (1.5)$$

$$k_{l,tot} = K_l + 2 \cdot k_{elbow} = 20.0 + 2 \cdot 0.3 = 20.6 \quad (1.6)$$

Moreover, the thermodynamic properties for the water (Cold Leg) and the steam (Hot Leg) are taken from Coolprop. In particular, the saturation temperature, the densities, specific enthalpies of dry saturated vapor and saturated liquid, and the viscosities are calculated. As a result it is possible to evaluate the mass flow and the buoyancy force:

$$\dot{m} = \frac{P_{th}}{h_v - h_l} = 23.12 \quad [kg/s] \quad (1.7)$$

$$BF = (\rho_l - \rho_v) \cdot g = (739.72 - 36.53) \cdot 9.81 = 6898.29 \quad [N/m^3] \quad (1.8)$$

Before calculating the hydraulic losses, the physical dimensions of the pipes are derived from nominal data and converted to the International System of Units (SI). The internal diameter D_m is derived from the nominal pipe dimensions (Outside Diameter and Wall Thickness/STD). Since the input data is provided in inches, a conversion factor of 0.0254 is applied to obtain the value in meters:

$$D_m = (D_{out} - 2 \cdot t_{wall}) \cdot 0.0254 \quad (1.9)$$

Supposing a perfectly circular pipe cross-section, then the cross-sectional flow area A is determined:

$$A = \frac{\pi \cdot D_m^2}{4} \quad [m^2] \quad (1.10)$$

The flow parameters are calculated separately for the liquid phase (Cold Leg) and the steam phase (Hot Leg). The velocities are determined by the conservation of mass, so it is possible to estimate the dynamic pressures and the Reynolds numbers.

$$v_l = \frac{\dot{m}}{\rho_l \cdot A}, \quad v_v = \frac{\dot{m}}{\rho_v \cdot A} \quad [\text{m/s}] \quad (1.11)$$

$$p_l = \frac{1}{2} \rho_l v_l^2, \quad p_v = \frac{1}{2} \rho_v v_v^2 \quad [\text{Pa}] \quad (1.12)$$

$$Re_l = \frac{\rho_l \cdot v_l \cdot D_m}{\mu_l}, \quad Re_v = \frac{\rho_v \cdot v_v \cdot D_m}{\mu_v} \quad (1.13)$$

1.2.1 Friction Factor calculation

Depending on the Reynolds number (Re), the friction factor f is evaluated using different correlations for laminar ($Re < 3000$) and turbulent ($Re > 3000$) flow regimes. Smooth pipes must be considered for the point A and B of the assignment, instead rough pipes for the point C.

- **Smooth Pipes:**

$$f = \frac{64}{Re} \quad Re < 3000 \quad (1.14)$$

$$f = \frac{0.316}{Re^{0.25}} \quad Re > 3000 \quad (1.15)$$

- **Rough Pipes (Haaland Equation):** In this case, the relative roughness ϵ/D is considered. The code implements the Haaland correlation, which is an explicit approximation of the Colebrook-White equation for the turbulent regime:

$$f = \frac{64}{Re} \quad Re < 3000 \quad (1.16)$$

$$\frac{1}{\sqrt{f}} = -1.8 \log_{10} \left(\left(\frac{\epsilon/D}{3.7} \right)^{1.11} + \frac{6.9}{Re} \right) \quad Re > 3000 \quad (1.17)$$

1.3 Calculations

- **Point A:** It is suggested to find an initial value of h ($h < L$ from the system layout), considering fixed L (the horizontal length) equal to 10 m and smooth pipes.

To evaluate h , the total pressure drops in the both branches of the circuit have to be calculated by summing the distributed losses along the piping and the concentrated (localized) losses due to components—such as the heater and cooler—and fittings.

$$\Delta p_{distr} = f_l \frac{L}{D_m} \frac{1}{2} \rho_l v_l^2 + f_v \frac{L}{D_m} \frac{1}{2} \rho_v v_v^2 \quad (1.18)$$

$$\Delta p_{loc} = K_{l,tot} \frac{1}{2} \rho_l v_l^2 + K_{v,tot} \frac{1}{2} \rho_v v_v^2 \quad (1.19)$$

$$\Delta p = \Delta p_{distr} + \Delta p_{loc} \quad (1.20)$$

Consequentially, h can be evaluated as follows:

$$h = \frac{\Delta p_{tot}}{BF} \quad (1.21)$$

The parametric results linking the standard pipe sizes to the corresponding loop heights obtained from the numerical model are organized in Table 1.2.

Table 1.2: Computed loop height (h) as a function of Nominal Pipe Size (NPS) and internal diameter (D_{int}) for Point A.

NPS	D_{int} [m]	Height h [m]
1/8"	0.0068	36 033 564.73
1/4"	0.0092	10 554 636.23
3/8"	0.0125	3 090 788.61
1/2"	0.0158	1 208 494.27
3/4"	0.0209	388 699.76
1"	0.0266	147 000.25
1 1/4"	0.0351	48 771.60
1 1/2"	0.0409	26 246.13
2"	0.0525	9 619.83
2 1/2"	0.0627	4 713.33
3"	0.0779	1 971.56
3 1/2"	0.0901	1 100.54
4"	0.1023	662.98
5"	0.1282	267.92
6"	0.1541	128.30
8"	0.2027	42.71
10"	0.2545	17.17
12"	0.3048	8.34
14"	0.3365	5.61
16"	0.3873	3.19
18"	0.4381	1.95
20"	0.4889	1.26
22"	0.5397	0.85
24"	0.5906	0.59
30"	0.7429	0.24
32"	0.7937	0.18
34"	0.8445	0.14
36"	0.8953	0.11
42"	1.0477	0.06

Analyzing the tabulated data, it is possible to notice that the height required for the NPS 12" pipe is equal to **8.34 m**, which is strictly less than the layout threshold $L = 10$ m (whereas the NPS 10" pipe requires an unfeasible height of 17.17 m). This confirms that the 12-inch pipe is the first configuration, starting from the smallest diameters, to be physically compatible with the design constraints.

1.4 Conclusions

The analysis conducted in this assignment validates the design of a passive decay heat removal system for a nuclear power plant, specifically sized to remove $34.8 \text{ MW}_{\text{th}}$ (representing 1% of the AP1000 thermal power). By applying the steady-state momentum balance where $\Delta p_{\text{buoyancy}} = \Delta p_{\text{loss}}$, the following engineering conclusions can be drawn:

- **System Feasibility and Sizing:** The **NPS 12"** pipe was identified as the smallest standard diameter capable of satisfying the physical constraints of the loop. In Point A, with a fixed horizontal length $L = 10 \text{ m}$, the required height $h = 8.34 \text{ m}$ is less than L , confirming the geometric viability of the rectangular loop layout.
- **Impact of Loop Geometry (Point B):** When the horizontal length was coupled to the vertical height ($L = h$), the required height decreased slightly to **8.33 m**. This demonstrates that minimizing the total pipe length reduces the distributed friction losses, thereby requiring a slightly lower buoyancy head to maintain the necessary natural circulation flow rate.
- **Influence of Pipe Roughness (Point C):** The transition from smooth pipes to a roughness of $5.0 \times 10^{-5} \text{ m}$ increased the required height to **8.37 m**. This result quantifies how increased surface friction, calculated via the Colebrook or Haaland correlations, necessitates a higher driving head (increased h) to overcome the additional pressure drops.
- **Reliability of Passive Mechanisms:** The results confirm that natural circulation can effectively serve as a driving force for safety systems without the need for external energy or active pumps. By relying on the density gradient between the hot leg (saturated steam) and the cold leg (saturated liquid) at 70 bar, the system ensures robust decay heat removal during accidental transients.

In summary, the NPS 12" configuration provides a robust and feasible design for the passive decay heat removal system, maintaining a balance between hydraulic efficiency and industrial standard compliance.

Chapter 2

Assignment 1.2 - Analysis of a Passive Decay Heat Removal System

2.1 Introduction

Natural circulation represents a cornerstone phenomenon in nuclear engineering, particularly for the development of passive safety features. A detailed thermal-hydraulic analysis of a passive Decay Heat Removal System (DHRS) is designed for a nuclear fission reactor. The objective of this assignment is to perform a formal verification of an existing design to ensure it remains compliant with safety requirements under accidental conditions.

The DHRS is engineered to provide core cooling and exhaust decay heat following a reactor shutdown. The system is divided into three interconnected circuits, as the Figure 2.1 :

- Primary System Circuit (PSC), which removes heat directly from the reactor core via natural circulation;
- Intermediate System Circuit (ISC) which receives heat from the PSC through the first heat exchanger (HX1);
- Tertiary System Circuit (TSC), which transfers heat from the second heat exchanger (HX2) to a cooling tower, where steam is condensed and the heat is exhausted to the environment.

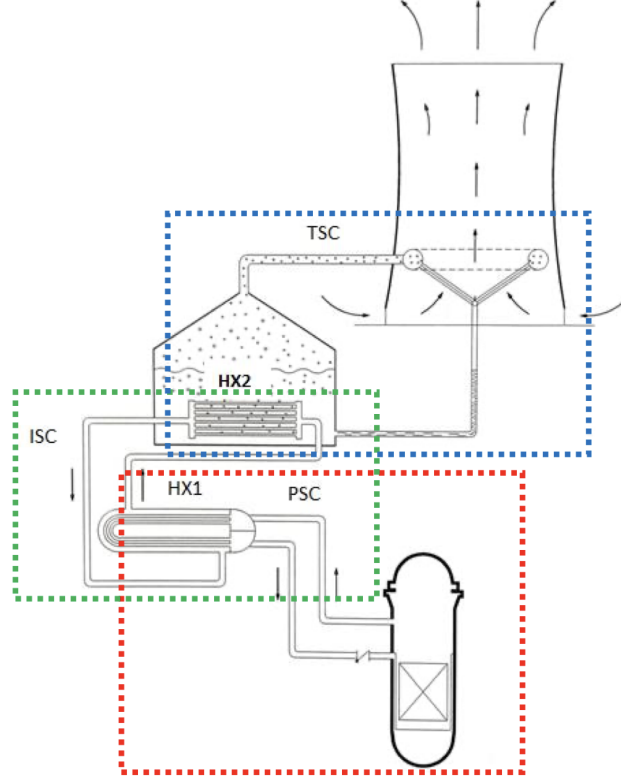


Figure 2.1: Scheme of the passive decay heat removal system.

During nominal operations, the DHRS remains inactive due to a check valve in the cold leg. In the event of a primary pump failure, the resulting pressure differential triggers the valve, initiating natural circulation driven by the density gradient between the cold downcomer and the heated core.

The goal of this assignment is to compute the mass flow rate and the coolant temperature in the PSC and ISC, starting from initial guesses.

2.1.1 Main Assumptions

To model the thermal-hydraulic behavior of the DHRS, the following simplifying assumptions have been adopted:

- **Steady-State Decay Power:** The system is assumed to be in a steady-state condition, neglecting the initial transient immediately following the reactor shutdown. While the core thermal power drops to approximately 7% of its nominal value ($P_{nom} = 600 MW_{th}$) right after shutdown, this analysis specifically considers the quasi-steady state reached after a few hours, where the decay heat to be removed is 1% of the nominal thermal power, resulting in a constant power level of $P = 6 MW_{th}$.
- **Ideal Heat Sink:** The cooling tower is assumed to exhaust all the power necessary to maintain the HX2 pool at constant saturation conditions (1 bar).
- **Constant Properties:** A constant coolant pressure is assumed within each of the three circuits to compute the thermophysical properties of water and steam.
- **Negligible fouling resistance:** No fouling resistance is considered within the heat exchangers.
- **HX1 Correction Factor:** A correction factor (F_T) of 0.7 is applied to the Mean Log Temperature Difference (LMTD) calculations for HX1, as it is not a pure countercurrent heat exchanger.

- **Vessel Pressure Losses:** The pressure drop in the vessel is modeled using a constant loss coefficient derived from a nominal $\Delta p_{vessel} = 1.2$ bar at a mass flow rate of 3200 kg/s .

2.1.2 Main Data

All the useful data to solve the problem are written in the Table 2.1

Table 2.1: Summary of Technical Data for DHRS Components

Component	Parameter	Symbol	Value	Unit
General	Nominal thermal power	P_{nom}	600	MW _{th}
	Vessel pressure drop ($\dot{m} \approx 3200 \text{ kg/s}$)	Δp_{vessel}	1.2	bar
PSC	Coolant pressure	p	75	bar
	Outer pipe diameter (Sch. 100)	D_{out}	16	in
	Pipe length (half)	L_{PSC}	8	m
	Pipe relative roughness	ϵ/D	$2 \cdot 10^{-4}$	-
	Pressure loss coefficient (90° bends)	k_{loss}	0.45	-
	Number of 90° bends	N_{bends}	4	-
	Valve pressure loss coefficient	$K_{loss, valve}$	0.12	-
	Elevation between hot and cold leg (H_1)	H_1	7	m
	Elevation top-bottom core (H_2)	H_2	3	m
HX1	Outer tube diameter	$D_{out, HX1}$	19.05	mm
	Tube thickness	t	1.24	mm
	Number of tubes	$N_{t, HX1}$	897	-
	Pitch	d_p	28.5	mm
	Shell inner diameter	D_{shell}	1.5	m
	Number of baffles	N_b	2	-
	Baffles spacing	$l_{baffles}$	1.6	m
	Average tube length	$L_{tubes, HX1}$	9.314	m
	Tube thermal conductivity	k_{th}	15	W/mK
	Headers average flow area	A_h	0.883	m ²
	Heat transfer area	$A_{ht, HX1}$	500	m ²
	Tube relative roughness	ϵ/D	10^{-4}	-
	Correction factor	F_T	0.7	-
ISC	Coolant pressure	p	70	bar
	Outer pipe diameter (Sch. 100)	D_{out}	16	in
	Pipe length (half)	L_{ISC}	20	m
	Pressure loss coefficient (90° bends)	k_{loss}	0.45	-
	Number of 90° bends	N_{bends}	2	-
	Tube relative roughness	ϵ/D	$2 \cdot 10^{-4}$	-
	Net elevation	H_{ISC}	10	m
HX2	Tube outer diameter	$D_{out, HX2}$	25.4	mm
	Tube thickness	t	1.24	mm
	Number of tubes	$N_{t, HX2}$	770	-
	Average tube length	$L_{tubes, HX2}$	7	m
	Manifold diameter	$D_{manifold}$	16	in
	Tube thermal conductivity	k_{th}	15	W/mK
	Heat transfer area	$A_{ht, HX2}$	430	m ²
	Tube relative roughness	ϵ/D	10^{-4}	-

2.2 Algorithm for the resolution

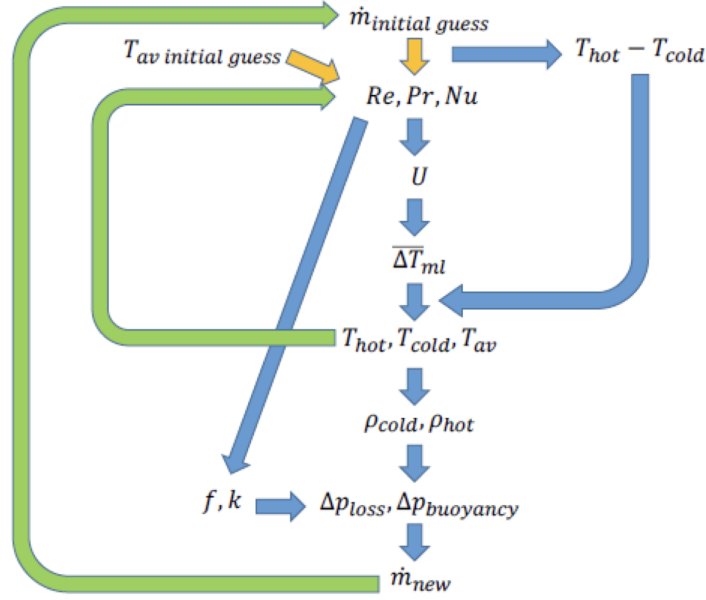


Figure 2.2: Diagram to understand the code setting.

This algorithm is valid for both ISC and PSC sides. There are two cycles, one within the other, where appropriate guesses should be done first. To ensure the accuracy of the thermal-hydraulic analysis, the thermophysical properties of the coolant (water) are dynamically evaluated using the CoolProp open-source library. The extraction process is centralized within a wrapper function to streamline the iterative calculations. The PropsSI method is utilized to retrieve the fundamental properties (density, dynamic viscosity, Prandtl number, specific heat, thermal conductivity) required for the momentum and energy balance equations. The function returns these values as a list, allowing the main algorithm to update fluid properties at each iteration based on the newly calculated average temperatures for the PSC and ISC circuits.

The algorithm begins with initial fundamental guesses for the mass flow rate and for the average temperature. As the mass flow rate ($\dot{m}$) and leg temperatures (T_{hot} , T_{cold}) are refined, the properties are re-evaluated at the new average temperature (T_{av}) until the convergence tolerance of 10^{-10} is met.

Moreover, it is important to consider that the system must remove decay power under quasi-steady-state conditions, set at 1% of the nominal thermal power:

$$\dot{Q} = P_{\text{nom}} \times 0.01 = 600 \cdot 10^6 \times 0.01 = 6 \text{ MW}_{\text{th}} \quad (2.1)$$

The hydraulic resistance of the vessel is calculated from the nominal data:

$$R_{\text{vessel}} = \frac{\Delta p_{\text{vessel, nom, Pa}}}{\dot{m}_{\text{vessel, nom}}^2} = \frac{1.2 \cdot 10^5 \text{ Pa}}{(3200 \text{ kg/s})^2} = 0.0117 \frac{\text{Pa}}{(\text{kg/s})^2} \quad (2.2)$$

For each duct, the internal diameter and flow area are evaluated as follows, taking into account that in heat exchangers it is necessary to multiply by the number of tubes N_t to obtain the total area:

$$D_{\text{in}} = D_{\text{out}} - 2t \quad (2.3)$$

$$A = \frac{\pi D_{\text{in}}^2}{4} \quad (2.4)$$

Moreover the total internal and external lateral surface areas must be considered for the heat exchangers:

$$A_{\text{lat,int}} = \pi \cdot L_{\text{tubes}} \cdot D_{\text{in}} \cdot N_t \quad (2.5)$$

$$A_{\text{lat,out}} = \pi \cdot L_{\text{tubes}} \cdot D_{\text{out}} \cdot N_t \quad (2.6)$$

For the HX1 heat exchanger, the flow area of the shell side and the equivalent diameter of the shell side for the triangular lattice are computed as:

$$D_{eq,shell} = \frac{2\sqrt{3}d_p^2}{\pi D_{tube}} - D_{tube} \quad (2.7)$$

$$A_{\text{shell}} = \frac{D_{\text{shell}}}{d_p} (d_p - D_{\text{tube}}) l_{\text{baffles}} \quad (2.8)$$

To calculate the pressure drops within the PSC and ISC, the Darcy friction factor (f) must be determined. It is achieved by distinguishing between different flow regimes, thanks to the Reynolds number, and applying the Haaland equation for turbulent conditions.

- Laminar Flow ($Re < 2300$): The friction factor is calculated using the exact analytical solution for circular pipes:

$$f = \frac{64}{Re} \quad (2.9)$$

- Turbulent Flow ($Re \geq 2300$): In this regime, the friction factor depends on both the Reynolds number and the relative roughness (ϵ/D). To avoid the iterative resolution of the implicit Colebrook-White equation, the script utilizes the explicit **Haaland** approximation, rearranged in the code to solve directly for f :

$$f = \left[-1.8 \log_{10} \left(\left(\frac{\epsilon/D}{3.7} \right)^{1.11} + \frac{6.9}{Re} \right) \right]^{-2} \quad (2.10)$$

2.3 Intermediate System Circuit (ISC) resolution

The steady-state resolution of the Intermediate System Circuit (ISC) is achieved through a nested iterative algorithm. This process ensures that the mass flow rate is consistent with the thermal power balance and the hydraulic resistance of the circuit.

The initial guesses made for the ISC are the following:

- mass flow rate $\dot{m}_{guess} \approx 100 \text{ kg/s}$
- Average temperature: $T_{\text{av}} \approx 120^\circ\text{C}$

The temperature difference across the circuit, ΔT_{isc} is computed via a power balance and utilized to determine the leg temperatures through the heat exchanger characteristic equations.

$$\Delta T_{isc} = \frac{\dot{Q}}{\dot{m}_{guess} \cdot cp_{ISC}} \quad (2.11)$$

2.3.1 Convective heat transfer coefficients for the ISC coolant flowing inside the HX2 tubes

The second heat exchanger (HX2) is an in-pool type component where the heat transfer occurs between the ISC coolant and a large water volume. The ISC coolant flows through a bundle of straight tubes submerged inside the pool. To ensure a proper flow distribution, a manifold is present upstream and downstream of the tubes. These are designed to have the same area of the ISC main pipe. The water on the pool side is assumed in saturation conditions at 1 bar.

Dimensionless numbers for the HX2 can be evaluated to get the value of the internal convective heat exchange coefficient $h_{int,HX2}$ using the Dittus-Boelter correlation:

$$Re_{HX2} = \frac{4 \cdot \dot{m}_{guess}}{\mu_{ISC} \cdot \pi \cdot D_{in,HX2} \cdot N_t} \quad (2.12)$$

$$Nu_{HX2} = 0.023 \cdot Re_{HX2}^{0.8} \cdot Pr_{ISC}^{0.3} \quad (2.13)$$

$$h_{int,HX2} = \frac{Nu_{HX2} \cdot k_{ISC}}{D_{in,HX2}} \quad (2.14)$$

For the pool-side, the McAdams correlation is employed to evaluate the external heat transfer coefficient ($h_{out,HX2}$). This correlation is based on the heat flux through the HX tubes q'' and wall superheating ΔT_{sat} with respect to pool saturation temperature:

$$q'' = 2.257 \cdot \Delta T_{sat}^{3.86} \quad (2.15)$$

Therefore, from the inverse formula:

$$\Delta T_{sat} = \left(\frac{\dot{Q}}{2.257 \cdot A_{HX2}} \right)^{\frac{1}{3.86}} \quad (2.16)$$

$$h_{out,HX2} = \frac{\dot{Q}}{A_{HX2} \cdot \Delta T_{sat}} \quad (2.17)$$

2.3.2 Global heat transfer for HX2

The total thermal resistance ($R_{tot,HX2}$) accounts for internal convection, conduction through the tube wall, and external pool-side convection can be evaluated:

$$R_{tot,HX2} = \frac{1}{h_{int,HX2} A_{lat,int,HX2}} + \frac{\ln(D_{out,HX2}/D_{in,HX2})}{2\pi k_{th} N_t L_{tubes,HX2}} + \frac{1}{h_{out,HX2} A_{lat,out,HX2}} \quad (2.18)$$

The global heat transfer coefficient (U_{HX2}) is defined relative to the external area $A_{lat,out,HX2}$:

$$U_{HX2} = \frac{1}{R_{tot,HX2} \cdot A_{lat,out,HX2}} \quad (2.19)$$

The Log Mean Temperature Difference calculation (ΔT_{ml}) required to transfer the decay power is possible:

$$\Delta T_{ml} = \frac{\dot{Q}}{A_{ht,HX2} \cdot U_{HX2}}$$

Given that the heat sink (the pool) remains at a constant saturation temperature (T_{sat}), the ISC leg temperatures are determined iteratively. The cold leg temperature $T_{cold,ISC}$ is derived from the relationship for a constant temperature sink, and so the hot leg temperature $T_{hot,ISC}$ is evaluated accordingly:

$$T_{cold,ISC} = T_{sat} + \frac{\Delta T_{ISC}}{\exp\left(\frac{\Delta T_{ISC}}{\Delta T_{ml}}\right) - 1} \quad (2.20)$$

$$T_{hot,ISC} = T_{cold,ISC} + \Delta T_{ISC} \quad (2.21)$$

At this point, the loop updates the average temperature $T_{av} = (T_{hot} + T_{cold})/2$ and re-evaluates fluid properties until the relative change in T_{av} falls below the specified tolerance (10^{-10}).

2.3.3 Evaluation of localized and distributed losses

Once thermal convergence is achieved, the algorithm evaluates the hydraulic resistance of the circuit to update the mass flow rate. Thermophysical properties are updated for the hot and cold legs using the converged temperatures (T_{hot}, T_{cold}). The Reynolds numbers for the piping and the HX1 shell side are computed to determine the flow regime. The friction factors (f) are then calculated using the Haaland approximation for turbulent flow.

To complete the hydraulic analysis of the Intermediate System Circuit (ISC), the localized and distributed pressure loss components are studied.

1. Distributed Losses

They account for the fluid friction against the walls of the conduits over a given length. These are calculated as:

- **Hot Leg Resistance:**

$$\mathcal{R}_{\text{dist,h,ISC}} = \frac{f_{\text{hot}} \cdot L_{\text{ISC}}}{2 \cdot D_{\text{in,ISC}} \cdot \rho_{\text{hot,ISC}} \cdot A_{\text{ISC}}^2} \quad (2.22)$$

- **Cold Leg Resistance:**

$$\mathcal{R}_{\text{dist,c,ISC}} = \frac{f_{\text{cold}} \cdot L_{\text{ISC}}}{2 \cdot D_{\text{in,ISC}} \cdot \rho_{\text{cold,ISC}} \cdot A_{\text{ISC}}^2} \quad (2.23)$$

- **HX2 Tube Resistance:**

$$\mathcal{R}_{\text{dist,HX2}} = \frac{f_{\text{HX2}} \cdot L_{\text{tubes,HX2}}}{2 \cdot D_{\text{in,HX2}} \cdot \rho_{\text{ISC}} \cdot A_{\text{tot,HX2}}^2} \quad (2.24)$$

2. Localized losses

Localized losses represent pressure drops due to changes in flow direction, geometry, or obstacles. The first step involves defining the dimensionless localized losses coefficients for each singularity in the circuit.

- **90° bends:** The total resistance for the piping elbows is determined by the number of bends per leg and their individual loss coefficient ($k_{\text{loss}} = 0.45$):

$$K_{\text{ISC,elb}} = N_{\text{bends}} \cdot k_{\text{loss}} \quad (2.25)$$

- **HX2 manifold:** Sudden enlargement and contraction occur at the entrance and exit of the HX2 tube bundle. These coefficients are calculated based on the area ratio between the ISC pipe (A_{ISC}) and the total tube area ($A_{\text{tot,HX2}}$):

$$K_{\text{enl,HX2}} = \left(1 - \frac{A_{\text{ISC}}}{A_{\text{tot,HX2}}}\right)^2, \quad K_{\text{cont,HX2}} = 0.5 \cdot \left(1 - \frac{A_{\text{ISC}}}{A_{\text{tot,HX2}}}\right) \quad (2.26)$$

- **HX1 shell-side Baffles:** The complex flow path across the tube bundle and baffles on the primary side interface is modeled using a specific correlation:

$$k_{\text{shell}} = 8 \cdot \frac{0.227}{Re_{\text{shell}}^{0.193}} \cdot \frac{D_{\text{shell}}}{D_{\text{eq,shell}}} \cdot (N_{\text{baffle}} + 1) \quad (2.27)$$

Once the coefficients are established, they are converted into hydraulic resistances within the numerical loop. This conversion accounts for the local fluid density and the specific flow area of the component:

- **Piping elbows resistance:** The resistances for the hot and cold legs are calculated using the respective local densities:

$$\mathcal{R}_{\text{loc,elb,h,ISC}} = \frac{K_{\text{ISC,elb}}}{2\rho_{\text{hot,ISC}}A_{\text{ISC}}^2}, \quad \mathcal{R}_{\text{loc,elb,c,ISC}} = \frac{K_{\text{ISC,elb}}}{2\rho_{\text{cold,ISC}}A_{\text{ISC}}^2} \quad (2.28)$$

- **HX2 inlet manifold expansion resistance and HX2 outlet manifold contraction resistance:** The expansion loss is evaluated at the hot leg density, while the contraction loss is evaluated at the cold leg density:

$$\mathcal{R}_{\text{loc,enl,HX2}} = \frac{K_{\text{enl,HX2}}}{2\rho_{\text{hot,ISC}}A_{\text{ISC}}^2}, \quad \mathcal{R}_{\text{loc,cont,HX2}} = \frac{K_{\text{cont,HX2}}}{2\rho_{\text{cold,ISC}}A_{\text{ISC}}^2} \quad (2.29)$$

- **HX1 shell side resistance:** The shell-side resistance utilizes the shell-side flow area and the average fluid density:

$$\mathcal{R}_{\text{loc,shell}} = \frac{k_{\text{shell}}}{2\rho_{\text{ISC}}A_{\text{shell}}^2} \quad (2.30)$$

2.3.4 Mass flow rate determination for the ISC

The steady-state mass flow rate within the Intermediate System Circuit (ISC) is determined by the equilibrium between the buoyancy driving head and the total hydraulic resistance of the loop. In a natural circulation regime, flow is driven by the density difference between cold and hot legs over the net elevation $H_{\text{ISC}} = 10$ m.

At steady-state, this buoyancy head must exactly balance the total pressure losses, expressed as follows:

$$\Delta p_{\text{buoyancy}} = \Delta p_{\text{loss}} \implies (\rho_{\text{cold,ISC}} - \rho_{\text{hot,ISC}}) \cdot g \cdot H_{\text{ISC}} = \mathcal{R}_{\text{tot,ISC}} \cdot \dot{m}^2 \quad (2.31)$$

In this formulation, $\mathcal{R}_{\text{tot,ISC}}$ represents the global hydraulic impedance of the circuit. This term aggregates all distributed and localized pressure drop contributions into a single resistance parameter:

$$\mathcal{R}_{\text{tot,ISC}} = \mathcal{R}_{\text{dist,h}} + \mathcal{R}_{\text{dist,c}} + \mathcal{R}_{\text{dist,HX2}} + \mathcal{R}_{\text{loc,elb,h}} + \mathcal{R}_{\text{loc,elb,c}} + \mathcal{R}_{\text{loc,enl,HX2}} + \mathcal{R}_{\text{loc,cont,HX2}} + \mathcal{R}_{\text{loc,shell}} \quad (2.32)$$

By rearranging the momentum balance, the numerical script iteratively updates the mass flow rate ($\dot{m}_{\text{new}}$) until the relative error between iterations falls below the specified tolerance (10^{-10}):

$$\dot{m}_{\text{new}} = \sqrt{\frac{(\rho_{\text{cold,ISC}} - \rho_{\text{hot,ISC}}) \cdot g \cdot H_{\text{ISC}}}{\mathcal{R}_{\text{tot,ISC}}}} \quad (2.33)$$

This approach ensures that the calculated mass flow rate ($\dot{m} \approx 70.61$ kg/s) is physically consistent with both the thermal power balance and the specific geometry of the piping and heat exchangers.

2.3.5 Final results for the ISC

The following table summarizes the thermal-hydraulic performance of the Integrated System Component (ISC), including mass flow rates, temperature levels, and a detailed breakdown of pressure drops, which are evaluated using the Python code in Appendix 2.

Table 2.2: ISC system operational data and pressure drop breakdown.

Parameter	Value	Unit
Mass flow rate ($\dot{m}$)	70.61	kg/s
Hot leg temperature (T_{hot})	129.18	°C
Cold leg temperature (T_{cold})	109.07	°C
Pressure Drops (Δp)		Pa
Hot leg total	470.78	
– Distributed	224.20	
– Localized	246.58	
Cold leg total	464.87	
– Distributed	222.41	
– Localized	242.46	
HX2 (In-pool) total	425.89	
– Distributed	202.43	
– Localized	223.46	
HX1 (Shell side) total	203.26	
Total ISC Pressure Drop	1564.80	Pa

2.4 Primary System Circuit (PSC) resolution

The steady-state resolution of the Primary System Circuit (PSC) is performed using a nested iterative algorithm similar to the one implemented for the ISC.

In this case, the initial guesses made for the PSC are the following:

- mass flow rate $\dot{m}_{\text{guess}} \approx 80 \text{ kg/s}$
- Average temperature: $T_{\text{av}} \approx 120 \text{ °C}$

For every iteration of the inner loop, the thermophysical properties of the water (density, viscosity, Prandtl number, specific heat, and thermal conductivity) are updated based on the current average temperature and a constant pressure of 75 bar.

The temperature difference is computed by the thermal power balance:

$$\Delta T_{PSC} = \frac{\dot{Q}}{\dot{m}_{\text{guess}} \cdot cp_{PSC}} \quad (2.34)$$

2.4.1 Convective heat transfer coefficients for the PSC and ISC coolants flowing inside the HX1 tubes

For the thermal analysis of HX1, which is a countercurrent, shell and tubes HX with U tubes, with a triangular lattice, a similar approach to HX2 is followed, but accounting for the more complex geometry and the exchange between the Primary System Circuit (PSC) and the Intermediate System Circuit (ISC).

The convective heat transfer coefficient for the primary coolant flowing within the HX1 tubes is determined by evaluating the flow regime. The following dimensionless parameters are calculated, applying the Dittus-

Boelter correlation to the PSC fluid:

$$Re_{HX1} = \frac{4 \cdot \dot{m}_{\text{guess}}}{\mu_{\text{PSC}} \cdot \pi \cdot D_{\text{in},HX1} \cdot N_{t,HX1}} \quad (2.35)$$

$$Nu_{HX1} = 0.023 \cdot Re_{HX1}^{0.8} \cdot Pr_{\text{PSC}}^{0.3} \quad (2.36)$$

$$h_{\text{int},HX1} = \frac{Nu_{HX1} \cdot k_{\text{PSC}}}{D_{\text{in},HX1}} \quad (2.37)$$

For the shell-side (ISC), a specific correlation to compute the heat transfer coefficient is reported:

$$Pr_{\text{shell}} = \frac{\mu_{\text{ISC}} \cdot c_{p,\text{ISC}}}{k_{\text{ISC}}} \quad (2.38)$$

$$h_{\text{out},HX1} = 0.351 \cdot Re_{\text{shell}}^{0.55} \cdot \frac{k_{\text{th}}}{D_{\text{eq,shell}}} \cdot Pr_{\text{shell}}^{1/3} \quad (2.39)$$

2.4.2 Global heat transfer for HX1

To determine the heat exchange performance between the Primary System Circuit (PSC) and the Intermediate System Circuit (ISC), a global heat transfer analysis is performed. This involves calculating the overall thermal resistance and solving the temperature field through an iterative approach. The total thermal resistance ($R_{\text{tot},HX1}$) is modeled as a series of resistances, including internal convection (tube-side), conduction through the tube wall, and external convection (shell-side). In accordance with the problem assumptions, fouling resistances are neglected. The total resistance is expressed as:

$$R_{\text{tot},HX1} = \frac{1}{h_{\text{int},HX1} A_{\text{lat,int},HX1}} + \frac{\ln\left(\frac{D_{\text{out},HX1}}{D_{\text{in},HX1}}\right)}{2\pi k_{\text{th}} N_{t,HX1} L_{\text{tubes},HX1}} + \frac{1}{h_{\text{out},HX1} A_{\text{lat,out},HX1}} \quad (2.40)$$

The global heat transfer coefficient (U_{HX1}), referenced to the external tube surface area, is then derived:

$$U_{HX1} = \frac{1}{R_{\text{tot},HX1} \cdot A_{\text{lat,out},HX1}} \quad (2.41)$$

Because HX1 utilizes a shell-and-tube configuration with U-tubes rather than a pure countercurrent flow, the Logarithmic Mean Temperature Difference must be adjusted. A correction factor $F_T=0.7$ is applied to the calculation of the required ΔT_{ml} to account for the non-ideal flow arrangement:

$$\Delta T_{\text{ml}} = \frac{\dot{Q}}{A_{\text{ht},HX1} \cdot U_{HX1} \cdot F_T} \quad (2.42)$$

To resolve the steady-state temperatures of the PSC ($T_{\text{hot},PSC}$ and $T_{\text{cold},PSC}$), an analytical approach is implemented based on the heat balance between the primary and intermediate circuits. An exponential parameter, ψ , is defined to relate the temperature differences of both circuits to the corrected Logarithmic Mean Temperature Difference (ΔT_{ml}) of HX1:

$$\psi = \exp\left(\frac{\Delta T_{PSC} - \Delta T_{ISC}}{\Delta T_{\text{ml}}}\right) \quad (2.43)$$

Using this factor, the primary cold leg temperature is calculated as follows:

$$T_{\text{cold},PSC} = \frac{\psi \cdot T_{\text{cold},ISC} + \Delta T_{PSC} - T_{\text{hot},ISC}}{\psi - 1} \quad (2.44)$$

The hot leg temperature is subsequently determined as:

$$T_{\text{hot},PSC} = T_{\text{cold},PSC} + \Delta T_{PSC} \quad (2.45)$$

These values allow for the update of the average temperature used to compute thermophysical properties in the subsequent iteration.

2.4.3 Driving force

The driving force ($\Delta P_{driving}$) that sustains natural circulation is generated by the density difference between the cold and hot legs along the system's elevation. The PSC net elevation is divided into two parts: H_1 (7 m), representing the elevation between the hot and cold leg furthest points, and H_2 (3 m), representing the elevation difference in the core.

The density in the core varies from the bottom (highest density, $\rho_{cold,PSC}$) to the top (lowest density, $\rho_{hot,PSC}$), so an integration is needed to compute the contribution to the driving force from the density difference in the core.

$$\Delta P_{driving} = g \left[(\rho_{cold,PSC} - \rho_{hot,PSC})H_1 + \int_0^{H_2} (\rho_{cold,PSC} - \rho(z)) dz \right] \quad (2.46)$$

Following the assumption that the density changes linearly with the height of the core:

$$\rho(z) = \rho_{cold,PSC} + \frac{\rho_{hot,PSC} - \rho_{cold,PSC}}{H_2} z \quad (2.47)$$

The integral is resolved by considering the average density within the core, which simplifies the contribution of the H_2 term to the total driving force:

$$\int_0^{H_2} (\rho_{cold,PSC} - \rho(z)) dz = (\rho_{cold,PSC} - \rho_{core})H_2 \quad (2.48)$$

This modeling approach ensures that the calculated mass flow rate is consistent with the buoyancy available to overcome the pressure drops of the vessel and heat exchangers, thereby verifying the safety requirements of the DHRS design.

2.4.4 Evaluation of localized and distributed losses

Once the thermal field of the Primary System Circuit (PSC) is determined, the algorithm proceeds to evaluate the thermophysical properties and the hydraulic performance of the loop. This step is essential to quantify the pressure drops that the natural circulation driving force must overcome. Thermophysical properties are extracted for both the hot and cold legs using the calculated temperatures ($T_{hot,PSC}, T_{cold,PSC}$) at the constant system pressure. A key parameter for the momentum balance is the average density within the reactor core ($\bar{\rho}_{core}$). Consistent with the assumption of a linear temperature (and thus density) variation along the core height H_2 , the core density is computed as the arithmetic mean of the hot and cold leg densities:

$$\bar{\rho}_{core} = \frac{\rho_{hot} + \rho_{cold}}{2} \quad (2.49)$$

To characterize the flow regime and determine the distributed pressure losses, the Reynolds numbers (Re) are calculated for the hot and cold legs of the piping, as well as for the tube side of HX1. Then, the friction factors (f) for the various sections of the PSC are determined using the Haaland approximation for turbulent flow. By calculating separate friction factors for the hot and cold sections, the model accurately reflects the changes in hydraulic resistance caused by temperature-dependent viscosity variations. These coefficients are subsequently used to compute the total distributed and localized pressure losses within the PSC.

1. Distributed losses

- **Hot Leg Resistance:**

$$\mathcal{R}_{dist,h,PSC} = \frac{f_{hot,PSC} \cdot L_{PSC}}{2 \cdot D_{in,PSC} \cdot \rho_{hot,PSC} \cdot A_{PSC}^2} \quad (2.50)$$

- **Cold Leg Resistance:**

$$\mathcal{R}_{dist,c,PSC} = \frac{f_{cold,PSC} \cdot L_{PSC}}{2 \cdot D_{in,PSC} \cdot \rho_{cold,PSC} \cdot A_{ISC}^2} \quad (2.51)$$

- **HX1 Tube Resistance:**

$$\mathcal{R}_{\text{dist,HX1}} = \frac{f_{\text{HX1}} \cdot L_{\text{tubes,HX1}}}{2 \cdot D_{\text{in,HX1}} \cdot \rho_{\text{PSC}} \cdot A_{\text{tot,HX1}}^2} \quad (2.52)$$

2. Localized losses

Firstly, the localized losses coefficients are determined:

- **Piping Bends:** Total loss coefficient for the bends in each leg (2 elbows per leg):

$$K_{\text{PSC,elb,leg}} = N_{\text{bends,PSC}} \cdot k_{\text{loss}} \quad (2.53)$$

- **HX1 U-Tube Bends:**

$$K_{\text{elb,HX1}} = N_{\text{bends,HX1}} \cdot k_{\text{loss}} \quad (2.54)$$

- **HX1 Entrance Interfaces:** Expansion and contraction coefficients at the first interface of the heat exchanger:

$$K_{\text{enl1,HX1}} = \left(1 - \frac{A_{\text{PSC}}}{A_{\text{h}}}\right)^2, \quad K_{\text{cont1,HX1}} = 0.5 \cdot \left(1 - \frac{A_{\text{tot,HX1}}}{A_{\text{h}}}\right) \quad (2.55)$$

- **HX1 Exit Interfaces:** Expansion and contraction coefficients at the second interface (exit) of the heat exchanger:

$$K_{\text{enl2,HX1}} = \left(1 - \frac{A_{\text{tot,HX1}}}{A_{\text{h}}}\right)^2, \quad K_{\text{cont2,HX1}} = 0.5 \cdot \left(1 - \frac{A_{\text{PSC}}}{A_{\text{h}}}\right) \quad (2.56)$$

Once the coefficients are established, they are converted into hydraulic resistances:

- **Piping elbows resistance:**

$$\mathcal{R}_{\text{loc,elb,h,PSC}} = \frac{K_{\text{PSC,elb}}}{2\rho_{\text{hot,PSC}}A_{\text{PSC}}^2}, \quad \mathcal{R}_{\text{loc,elb,c,PSC}} = \frac{K_{\text{PSC,elb}}}{2\rho_{\text{cold,PSC}}A_{\text{PSC}}^2} \quad (2.57)$$

- **HX1 U-tube bends resistance:**

$$\mathcal{R}_{\text{loc,elb,HX1}} = \frac{K_{\text{elb,HX1}}}{2 \cdot \rho_{\text{PSC}} \cdot A_{\text{tot,HX1}}^2} \quad (2.58)$$

- **HX1 Entrance Transition Resistances:**

$$\mathcal{R}_{\text{loc,enl1,HX1}} = \frac{K_{\text{enl1,HX1}}}{2\rho_{\text{hot,PSC}}A_{\text{PSC}}^2}, \quad \mathcal{R}_{\text{loc,cont1,HX1}} = \frac{K_{\text{cont1,HX1}}}{2\rho_{\text{hot,PSC}}A_{\text{tot,HX1}}^2} \quad (2.59)$$

- **HX1 Exit Transition Resistances:**

$$\mathcal{R}_{\text{loc,enl2,HX1}} = \frac{K_{\text{enl2,HX1}}}{2\rho_{\text{cold,PSC}}A_{\text{tot,HX1}}^2}, \quad \mathcal{R}_{\text{loc,cont2,HX1}} = \frac{K_{\text{cont2,HX1}}}{2\rho_{\text{cold,PSC}}A_{\text{PSC}}^2} \quad (2.60)$$

- **PSC Check Valve Resistance:**

$$\mathcal{R}_{\text{loc,valve,PSC}} = \frac{K_{\text{loss,valve}}}{2\rho_{\text{cold,PSC}}A_{\text{PSC}}^2} \quad (2.61)$$

2.4.5 Mass flow rate determination for the PSC

The ultimate goal of the hydraulic analysis is to compute the mass flow rate in the PSC. This is achieved by balancing the natural circulation driving force against the total hydraulic resistance of the circuit. The total resistance of the PSC $\mathcal{R}_{\text{tot,PSC}}$ is defined as the summation of all distributed and localized resistance components identified in the preceding analysis, including the pressure drop across the vessel ($\mathcal{R}_{\text{vessel}}$). So, it can be expressed as:

$$\mathcal{R}_{\text{tot,PSC}} = \sum \mathcal{R}_{\text{dist}} + \sum \mathcal{R}_{\text{loc}} + \mathcal{R}_{\text{vessel}} \quad (2.62)$$

In steady-state conditions, the natural circulation mass flow rate is determined by the balance between the buoyancy-induced driving force and the total hydraulic pressure losses. The driving force, $\Delta P_{\text{driving}}$, is calculated by integrating the density differences along the circuit elevations H_1 and H_2 . Assuming a linear density variation within the core, the driving force simplifies to a function of the cold and hot leg densities ($\rho_{\text{cold,PSC}}, \rho_{\text{hot,PSC}}$) and the effective thermal centers:

$$\Delta P_{\text{driving}} = g(\rho_{\text{cold,PSC}} - \rho_{\text{hot,PSC}}) \left(H_1 + \frac{H_2}{2} \right) \quad (2.63)$$

By equating this driving force to the product of the total resistance and the square of the mass flow rate ($\Delta P_{\text{driving}} = \mathcal{R}_{\text{tot,PSC}} \cdot \dot{m}^2$), the mass flow rate is updated iteratively according to the following relation:

$$\dot{m}_{\text{new}} = \sqrt{\frac{g(\rho_{\text{cold,PSC}} - \rho_{\text{hot,PSC}}) \left(H_1 + \frac{H_2}{2} \right)}{\mathcal{R}_{\text{tot,PSC}}}} \quad (2.64)$$

So the value at which the mass flow rate converges is ($\dot{m} \approx 57.09$ kg/s) also reasonable as in ISC

2.4.6 Final results for the PSC

The following table details the steady-state thermal-hydraulic parameters of the Primary System Circuit (PSC), obtained via the Python algorithm in Appendix 2

Table 2.3: PSC system operational data and pressure drop breakdown.

Parameter	Value	Unit
Mass flow rate ($\dot{m}$)	57.09	kg/s
Hot leg temperature (T_{hot})	149.88	°C
Cold leg temperature (T_{cold})	125.19	°C
Pressure Drops (Δp)		Pa
Hot leg total	388.62	
– Distributed	59.93	
– Localized	328.69	
Cold leg total	401.76	
– Distributed	59.14	
– Localized	342.62	
HX1 (Tube side) total	962.07	
– Distributed	649.93	
– Localized	312.15	
Pressure Vessel total	38.20	
Total PSC Pressure Drop	1790.65	Pa

2.5 Conclusions

The analysis of the passive Decay Heat Removal System (DHRS) demonstrates its critical role in maintaining plant safety following a reactor SCRAM. In such accidental transients, the immediate cessation of the primary coolant pumps triggers the opening of the check valve, allowing natural circulation to develop due to the density gradient between the cold fluid in the downcomer and the hot fluid in the core. Without an effective DHRS, the accumulation of decay heat—which reaches a quasi-steady state of approximately 1% of nominal power (6 MW_{th}) shortly after shutdown—would lead to a dangerous temperature rise and potential core meltdown.

Based on the numerical simulation of the coupled Primary System Circuit (PSC) and Intermediate System Circuit (ISC), the following performance parameters were obtained:

- **Mass Flow Rates:** The system establishes a stable natural circulation flow of **57.09 kg/s** in the PSC and **70.61 kg/s** in the ISC.
- **Thermal Margins:** Under operational pressures of **75 bar** for the PSC and **70 bar** for the ISC, the respective saturation temperatures are approximately 290°C and 285°C .
- **System Stability:** The total pressure drops calculated are of the order of 10^{-2} bar, which are negligible with respect to the variation of the saturation temperature.

The results show that the maximum temperature outputs in both loops do not exceed 150°C . This provides a substantial safety margin of more than 135°C below the boiling point. By strictly maintaining the coolant as a single-phase liquid, the design effectively prevents the formation of steam pockets that could compromise heat transfer efficiency or induce mechanical stress.

Ultimately, this verification confirms that the DHRS is well-designed and capable of exhausting the decay heat to the Tertiary System Circuit (TSC) and the environment. Ensuring single-phase operation is critical not only for thermal-hydraulic stability but also as a primary safety barrier to prevent the risk of radioactive leakage, thereby keeping the reactor in a guaranteed safe state.

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